National Qualifications 2018

X707/77/11

Biology
Supplementary Sheet

TUESDAY, 15 MAY
9:00 AM – 11:30 AM

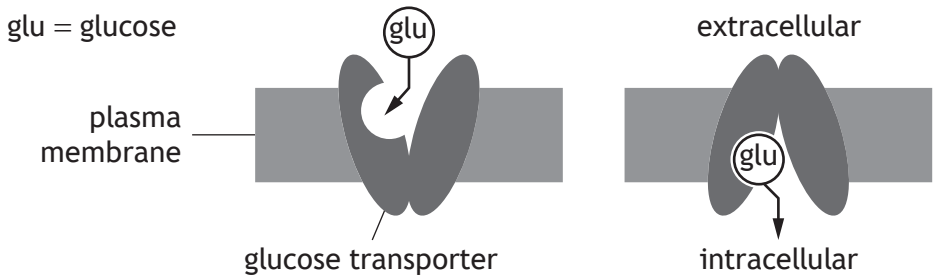
Supplementary sheet for question 1

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1. The diffusion of glucose across the plasma membrane of mammalian cells is facilitated by a family of related proteins called GLUT transporters. GLUT transporters have a specific binding site for glucose which alternately faces inside and outside the cell. The orientation of the binding site is brought about by a change in conformation as shown in Figure 1.

Figure 1

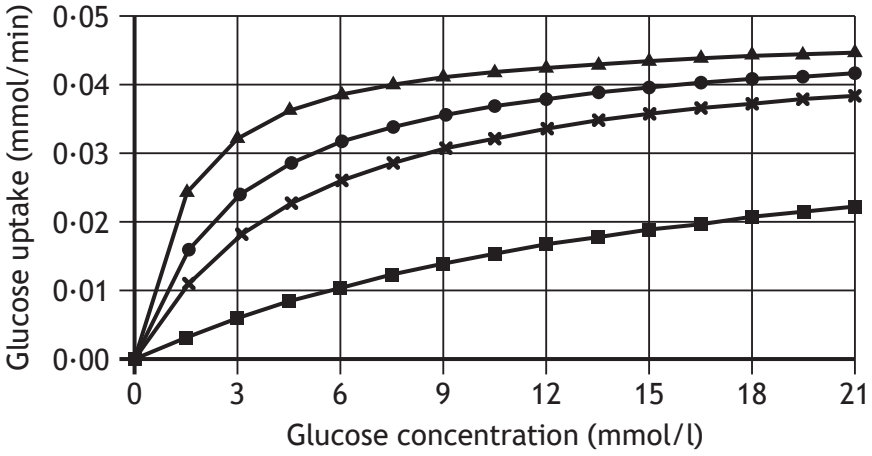


Several studies have measured the changes in rates of glucose uptake by GLUT transporters as the concentration of glucose is increased.

Results for four different GLUT transporters are shown in Figure 2.

For each type of GLUT transporter, the rate of transport levels off to a maximum value that is termed V_{max}. The glucose concentration at which the rate of transport is **half** V_{max} is defined as the K_M of the transporter. K_M values for four types of GLUT are shown in Figure 2.

Figure 2



Key

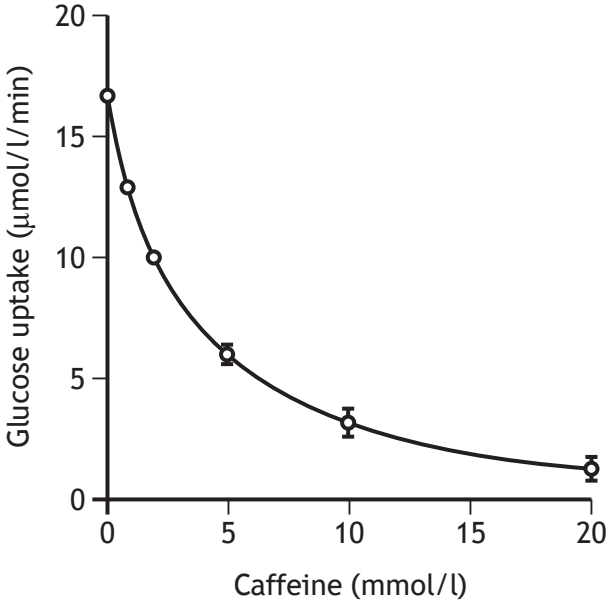
GLUT3 (K_M = 1·4 mmol/l)
GLUT1 (K_M = 3·0 mmol/l)
GLUT4 (K_M = 5·0 mmol/l)
GLUT2 (K_M = 17·0 mmol/l)

1. (continued)

Studies have suggested that the chemical caffeine is an inhibitor of glucose transport by GLUT1.

Figure 3 shows data obtained from a recent study of the effect of increasing caffeine concentration on the uptake of glucose by GLUT1. The uptake of glucose in these experiments was measured per litre of intracellular fluid.

Figure 3



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